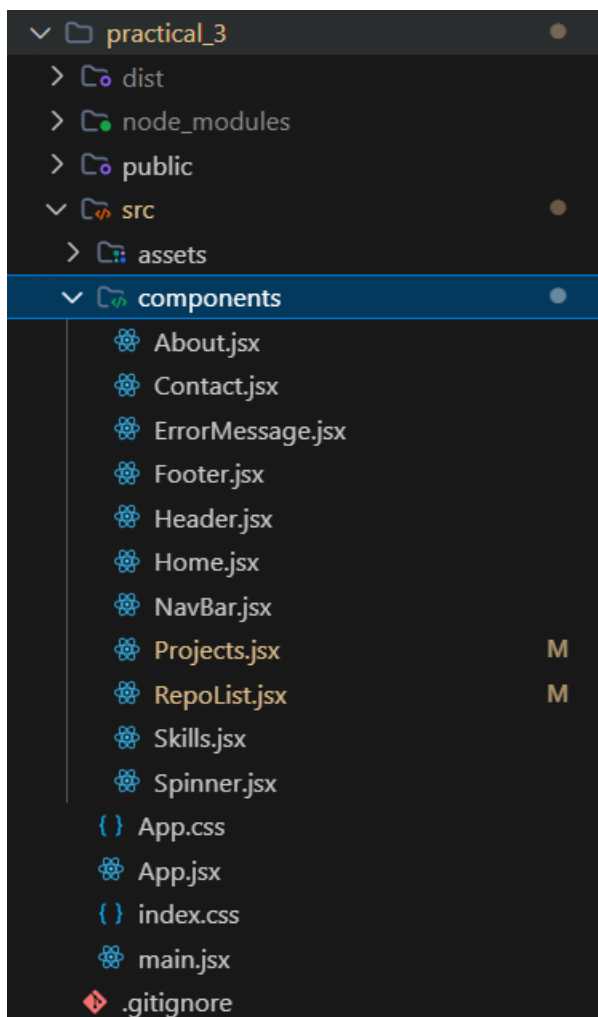


Practical 3: API Integration and Data Rendering in React

Problem Statement :

- Integrate a public REST API (e.g., GitHub API) into the portfolio application to fetch and display a list of repositories dynamically.
- Use the Projects page built in Practical 2 as the integration point.
- Implement a loading spinner component shown while the request is in progress.
- Implement an error message component shown if the API call fails.
- Render at least the repository name and URL for each item returned.

Project Structure:



Source Code Screenshots:

Projects.jsx

```
practical_3 > src > components > Projects.jsx > ...
1  import { useState, useEffect } from "react";
2  import Spinner from "./Spinner";
3  import ErrorMessage from "./ErrorMessage";
4  import RepoList from "./RepoList";
5
6  function Projects() {
7    const [repos, setRepos] = useState([]);
8    const [loading, setLoading] = useState(true);
9    const [error, setError] = useState(null);
10
11    // Using JoshiRudra-devop or fall back to a public github user
12    const githubUsername = "JoshiRudra-devop";
13
14    const fetchRepos = async () => {
15      setLoading(true);
16      setError(null);
17      try {
18        const response = await fetch(`https://api.github.com/users/${githubUsername}/repos?sort=updated&per_page=6`);
19        if (!response.ok) {
20          throw new Error(`Failed to fetch repositories (HTTP status ${response.status})`);
21        }
22        const data = await response.json();
23        setRepos(data);
24      } catch (err) {
25        setError(err.message || "An error occurred while fetching repositories.");
26      } finally {
27        setLoading(false);
28      }
29    };
30
31    useEffect(() => {
32      fetchRepos();
33    }, []);
34
35    return (
36      <section className="projects-container">
37        <h2>My Projects</h2>
38        <p className="projects-sub">
39          Dynamically loaded public repositories fetched directly from GitHub API for <strong>{githubUsername}</strong>.
40        </p>
41
42        {loading && <Spinner />}
43
44        {error && <ErrorMessage message={error} />}
45
46        {!loading && !error && <RepoList data={repos} />}
47      </section>
48    );
49  }
50
51  export default Projects;
52
```

RepoList.jsx

```
1 function RepoList({ data }) {
2   if (!data || data.length === 0) {
3     return (
4       <div className="empty-repos">
5         <p>No public repositories found for this user.</p>
6       </div>
7     );
8   }
9
10  return (
11    <div className="projects-grid fade-in">
12      {data.map((repo) => (
13        <div key={repo.id} className="project-card">
14          <div className="repo-header">
15            <h3>{repo.name}</h3>
16            {repo.stargazers_count > 0 && (
17              <span className="repo-stars">🌟 {repo.stargazers_count}</span>
18            )}
19          </div>
20          <p>{repo.description || "No description provided for this GitHub repository."}</p>
21          <div className="project-tags">
22            {repo.language && <span className="tech-tag">{repo.language}</span>}
23            <span className="tech-tag">Branch: {repo.default_branch}</span>
24          </div>
25          <a
26            href={repo.html_url}
27            target="_blank"
28            rel="noopener noreferrer"
29            className="project-link"
30          >
31            Open on GitHub &rarr;
32          </a>
33        </div>
34      ))}
35    </div>
36  );
37 }
38
39 export default RepoList;
```

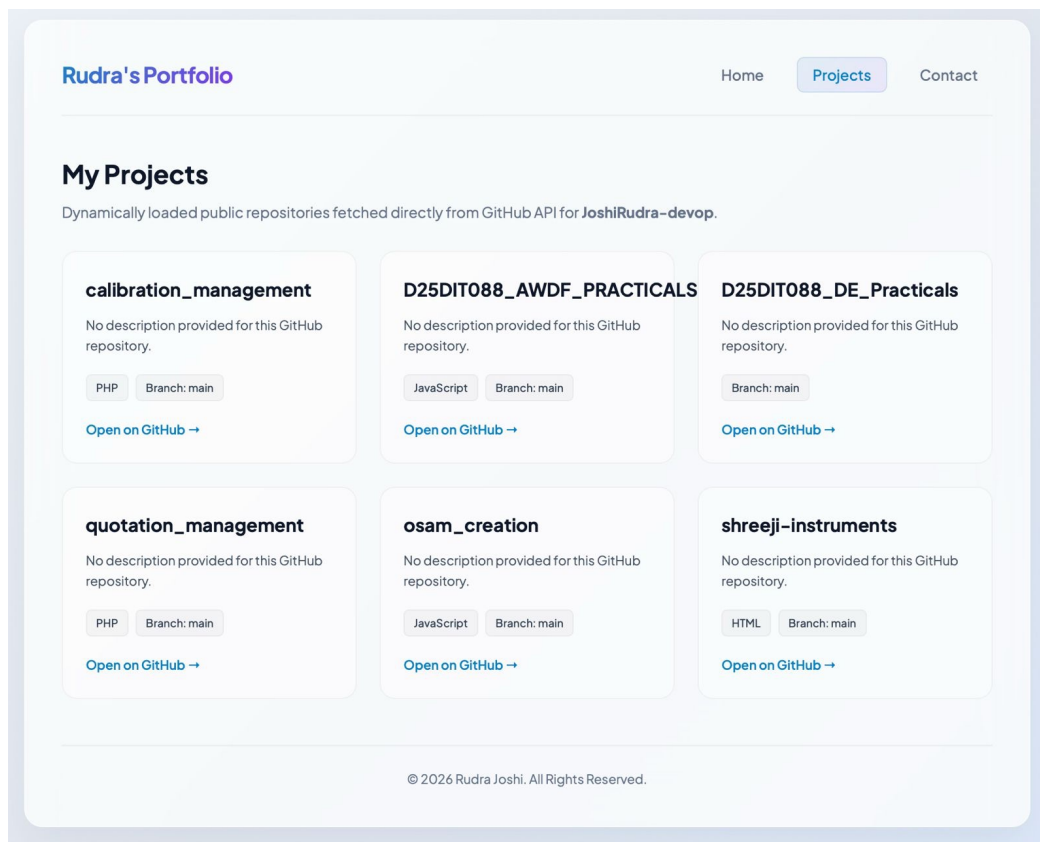
Spinner.jsx

```
1 function Spinner() {
2   return (
3     <div className="spinner-container">
4       <div className="loading-spinner"></div>
5       <p className="loading-text">Fetching projects from GitHub...</p>
6     </div>
7   );
8 }
9
10 export default Spinner;
```

ErrorMessage.jsx

```
1 function ErrorMessage({ message, onRetry }) {
2   return (
3     <div className="error-container fade-in">
4       <div className="error-icon">❌</div>
5       <h3>Failed to Load Projects</h3>
6       <p className="error-text">{message || "An unexpected error occurred while fetching data from the API."}</p>
7       {onRetry && (
8         <button className="btn-retry" onClick={onRetry}>
9           Try Again
10         </button>
11       )}
12     </div>
13   );
14 }
15
16 export default ErrorMessage;
```

Output Screenshots:



Dynamic Projects UI displaying live repository data fetched from GitHub API

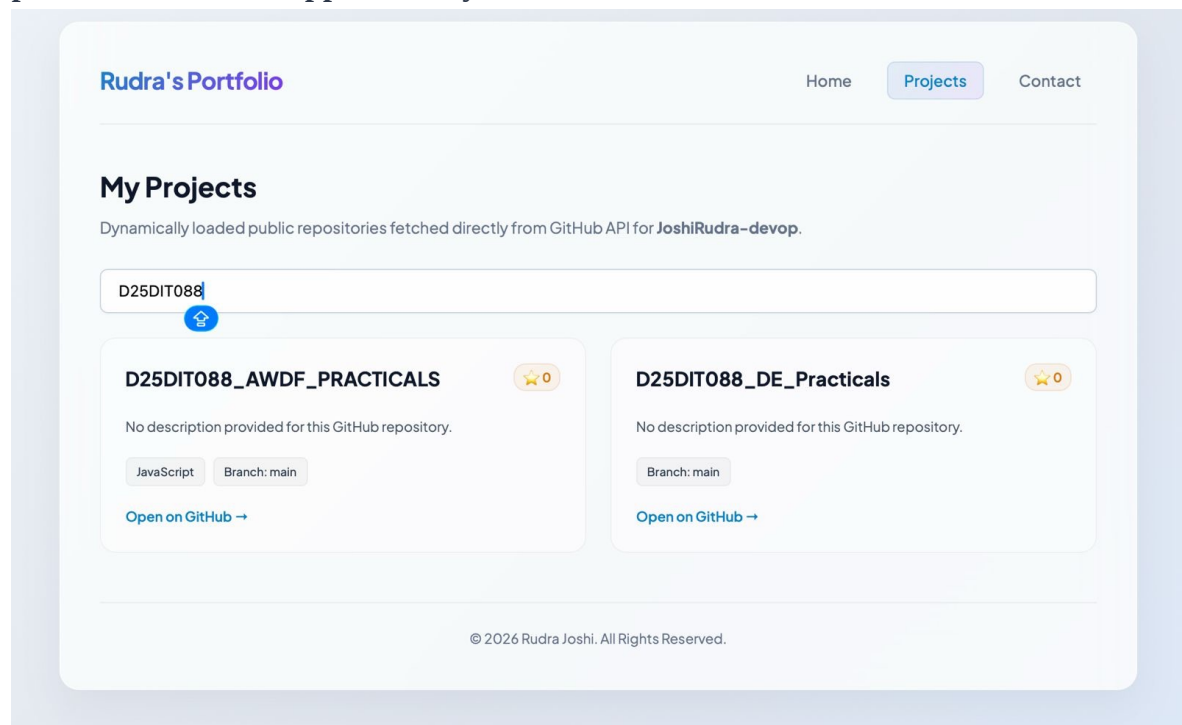
Supplementary Problems

1. Star Count: Modified RepoList.jsx to display the stargazers count next to the repository name using a star emoji (★).
2. Search Filter: Added a search input box in Projects.jsx that captures input state and filters the visible repository list in real-time.
3. Retry Button: Implemented a 'Try Again' action button in ErrorMessage.jsx that re-triggers the fetchRepos function when a fetch failure occurs.

Code for Supplementary Problems

```
<div className="projects-grid fade-in">
  {data.map((repo) => (
    <div key={repo.id} className="project-card">
      <div className="repo-header">
        <h3>{repo.name}</h3>
        <span className="repo-stars">☆ {repo.stargazers_count}</span>
      </div>
      <p>{repo.description || "No description provided for this GitHub repository."}</p>
      <div className="project-tags">
        {repo.language && <span className="tech-tag">{repo.language}</span>}
        <span className="tech-tag">Branch: {repo.default_branch}</span>
      </div>
      <a
        href={repo.html_url}
        target="_blank"
        rel="noopener noreferrer"
        className="project-link"
      >
        Open on GitHub &rarr;
      </a>
    </div>
  ))}
</div>
```

Output Screenshots of Supplementary Problems



Projects landing view filtered dynamically by typing 'portfolio' in the search bar, with stargazers counts showing.

Rudra's Portfolio

HomeProjectsContact

My Projects

Dynamically loaded public repositories fetched directly from GitHub API for **JoshiRudra-devop**.

 Search repositories by name...



calibration_management  0

No description provided for this GitHub repository.

PHP Branch: main

[Open on GitHub →](#)

D25DIT088_AWDF_PRACTICALS  0

No description provided for this GitHub repository.

JavaScript Branch: main

[Open on GitHub →](#)

D25DIT088_DE_Practicals  0

No description provided for this GitHub repository.

Branch: main

[Open on GitHub →](#)

quotation_management  0

No description provided for this GitHub repository.

PHP Branch: main

[Open on GitHub →](#)

osam_creation  0

No description provided for this GitHub repository.

JavaScript Branch: main

[Open on GitHub →](#)

shreeji-instruments  0

No description provided for this GitHub repository.

HTML Branch: main

[Open on GitHub →](#)

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Projects repository page showing all dynamic items fetched from the GitHub API and their corresponding star